FORUM: United Nations Office for Outer Space Affairs

ISSUE: Measures to Strengthen International Cooperation on Space Debris and Mega-Constellation Governance

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Introduction

The space debris crisis fundamentally threatens humanity's global satellite infrastructure and the accessibility of future space. Today, thousands of dead satellites, abandoned projectile remnants, and millions of untraceable micro-debris are orbiting around Earth at unimaginable velocity. This congestion heightens the risk of collisions, and collisions might cause structural damage to active satellites, resulting in problems in operational capabilities and disrupting data transmission used for terrestrial services. As a result, the likelihood of interruptions across global communication, weather forecasting, GPS systems, the stock market, and many other crucial things in modern society increases.

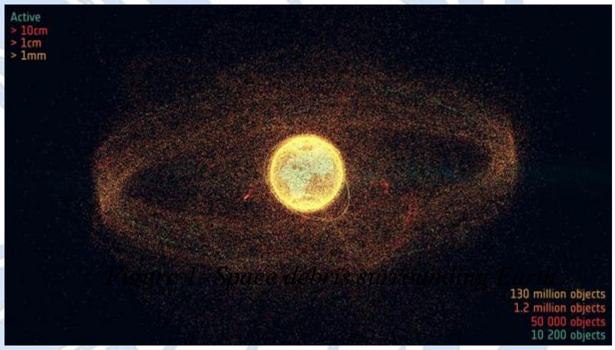

Figure 1: Space debris surrounding the Earth

Some nations and commercial companies are implementing strategies to mitigate this problem, such as deorbiting technology and collision-avoidance systems; however, their effectiveness is limited by disparate national standards, such as the U.S. Federal Communications Commission's (FCC) 5-year rule versus the Inter-Agency Space Debris Coordination Committee's (IADC) 25-year guideline: they are guidelines requiring to deorbit defunct Low Earth Orbit (LEO) satellites. Also, congestion at certain altitudes leads to inequality in nations' access to space, deepening friction regarding space resources. To guarantee a sustainable space environment, going beyond individual entities' fragmented responses, multilateral cooperation and international consensus are urgent.

Furthermore, the rapid increase in mega-constellations such as Starlink by SpaceX has transformed LEO, intensifying the need for orbital governance. To ensure sustainable access, traffic management and resource allocations, such as orbital space and radio frequency spectrum, for these large-scale satellite networks are required.

Background

Space exploration started in the 20th century. In the Cold War era, in the 1950s and 1960s, after World War II came to an end, the "space race" began, which was a political and technological competition between two superpowers, the United States and the Soviet Union. In that era, space activity was exclusive to a few governments and public agencies such as NASA, and they mostly focused on success in launching big, expensive satellites. Sputnik 1, launched by the Soviet Union in 1957, was the first satellite to enter Earth orbit. In 1961, the Thor-Ablestar rocket exploded and caused satellite breakup for the first time, generating more than 200 pieces of space debris. This event was the first major orbital breakup in history, marking the beginning of artificial space pollution by leaving long-lasting debris in high LEO. In response, space organizations began researching and implementing 'passivation', a process of removing internal stored energy, to prevent residual fuel explosions in orbit. In the 21st century, there have been a few debris-causing events, such as when China shot down its own weather satellite Fengyun-1C in 2007, and the United States's communication satellite Iridium 33 and Russia's dead military satellite Cosmos 2251 collided at about 11.7 kilometers per second, alarming scientists that space debris like dead satellites in this case needs caution. Also, this incident nearly doubled the tracked debris in LEO, compelling international bodies to establish formal space debris mitigation guidelines.

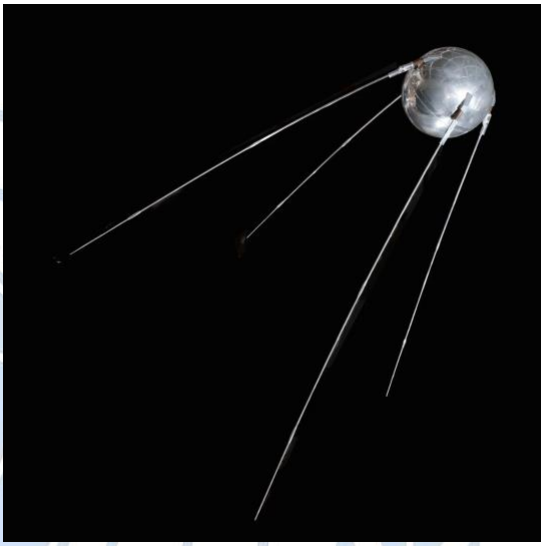

Figure 2: Sputnik 1

At present, the space industry is in a "New Space" era, defined by growing commercialization and a broader diversification of space activities. Rocket technology has evolved, making the recycling of rockets possible, leading to a decrease in the cost of launch, and small-scale satellite production technology has become popularized, enabling private companies to change space into a business stage that generates huge economic value. One of the most notable changes in the New Space era is the rapid increase in Mega-Constellations using LEO. To provide fast, uninterrupted internet everywhere on Earth, private companies, such as Starlink by SpaceX, Project Kuiper by Amazon, OneWeb, and governments are launching small satellites at an unprecedented rate in LEO. The number of active satellites operating in LEO has already exceeded the total number of satellites launched in the past few decades. An overwhelming number of satellites deployed at a certain altitude is engendering extreme space traffic saturation and increasing the danger of collision.

Problems Raised

The Cascading Risks of Kessler Syndrome

This sharp increase in congestion is heightening the risk of the pernicious scenario that threatens the sustainability of the space environment: "Kessler Syndrome." This concept was first introduced in a 1978 paper by NASA researchers Donald Kessler and Burton Cour-Palais.
It models cascading collisions, which occur when random collisions of artificial objects in orbit generate further debris. Kessler Syndrome is a long-term exponential risk that gradually exacerbates the orbital environment. Even tiny pieces of space debris travel at over 7 kilometers per second. A significant amount of energy is generated in collisions with satellites, creating numerous debris fragments, which in turn create more, and so on. The altitude range of 600-1000 km is especially vulnerable since atmospheric drag is too weak to provide natural self-cleansing in that range, allowing debris to persist for decades and centuries.

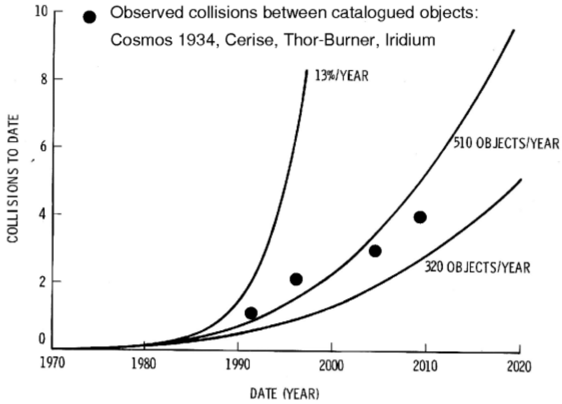

Figure 3: Mathematical modeling of cascading orbital collisions over time

The Kessler Syndrome might exponentially elevate the cost of global satellite infrastructure and the danger of structural damage, as it might impair satellite networks and high-precision Global Navigation Satellite Systems (GNSS). First, the microsecond-to-nanosecond time synchronization could compromise global power grids, increasing operational risks for critical infrastructure and creating the possibility of widespread power outages. At the same time, the timestamping of transactions across financial networks becomes impossible, causing immediate shutdowns of global financial systems and communications. Transportation logistics systems, including aviation, maritime, and rail, also become uncontrollable. The absence of precision agriculture and meteorological forecasting systems will deal a fatal blow even to global food supply chains and healthcare infrastructure.

Furthermore, Kessler Syndrome impacts terrestrial ecosystems and the global environment.
Not only does the risk of unmanaged large debris falling into densely populated areas materialize, but as countless pieces of space junk re-enter and burn up in the atmosphere, vast quantities of aluminum oxide particles are generated. The accumulation of aluminum oxide particles in the stratosphere catalyzes ozone layer depletion reactions and alters climate patterns, creating a severe cascading ripple effect where outer space congestion culminates in an environmental disaster for the entire terrestrial biosphere.

Deficits in International Space Law

Current international legal framework intended to regulate space debris has a clear gap. Current frameworks were designed for the government-led space development era and are insufficiently specific and enforceable, so they have clear limitations in effectively regulating private satellite operations and orbital congestion. For example, the 1972 Space Liability Convention stipulates the launching state's liability for fault-based damage. However, nonfunctional 'space debris' is not explicitly defined or separately regulated in the Convention, creating a legal ambiguity. Also, because liability under the Convention is primarily imposed on launching states rather than private entities, it is almost impossible to prove the fault of the other party or the entity responsible in the event of a collision caused by micro-debris that is difficult to track in Earth's orbit.

Eventually, current space treaties, such as the 1967 Outer Space Treaty (OST), are legally binding but lack sufficient specificity and enforcement mechanisms to manage large-scale monopolization of orbital resources or reduce space debris. Ultimately, the current lack of enforceable international guidelines and an integrated management interface create a hazardous orbital environment that severely compromises space accessibility for future generations.

International Actions

Expansion of the Space Situational Awareness (SSA) Data-Sharing Framework

To mitigate the risk of collision in orbit, the international community has focused on expanding the Space Situational Awareness (SSA) monitoring network and activating data sharing between multilateral entities. SSA is not a thing managed by a single transnational organization. Rather, it is a decentralized capability operated by individual national bodies. For example, government- and military-led agencies such as the U.S. Space Force's Combined Space Operations Center (CSpOC) and the European Union's EU SST use ground radar and sensor networks to track orbital objects. When a collision hazard is detected, these organizations issue Conjunction Data Messages (CDMs) to support governments and private satellite operators in taking evasive action. Through the 'Guidelines for the Long-term Sustainability (LTS) of Outer Space Activities' adopted by the UN Committee on the Peaceful Uses of Outer Space (COPUOS) in 2019, the committee formally recommended enhancing the accuracy of space object registration and fostering SSA data exchange through joint research.

Despite these actions, the global SSA system has faced security and technical limitations. First, SSA tracking possesses dual-use characteristics. It directly relates to military anti-satellite (ASAT) and reconnaissance capabilities, engendering a serious security dilemma. As such, major spacefaring nations, such as the United States, China, and Russia, are averse to sharing high-precision sensor data, apprehending the exposure of military surveillance capabilities and vulnerabilities, resulting in fragmentation issues where publicly disclosed data suffers from reduced accuracy or operational delays. Second, standardized data specifications and coordinate reference frames between various public agencies and private platforms are lacking. Such incongruity prevents large-scale mega-constellations like Starlink from seamlessly integrating real-time data into AI-based automated collision avoidance systems, ultimately leaving structural constraints that force a reliance on manual verification processes.

Active Debris Removal (ADR) Missions & Public-Private Partnerships

Space agencies are engaged in developing active debris removal (ADR) technology, which directly captures debris and burns it by making it re-enter the atmosphere, and practically applying it. Cooperation between public and private organizations is active, such as the ELSA-d mission led by Japanese private company Astroscale. ELSA-d was the first to prove the commercial possibility of ADR technology. It uses a magnetic-based mechanism to successfully separate and capture the target satellite in space. In addition, ClearSpace, also a private startup company, leads the ClearSpace-1 mission, which uses a robotic arm to capture and remove the dead VESPA rocket. The European Space Agency (ESA) signed a contract worth approximately 86 million euros with private startup ClearSpace, exemplifying a public budget-funded public-private partnership project. These projects were successful at demonstrating the technical viability of commercial ADR, establishing a practical foundation for market-driven orbital remediation.

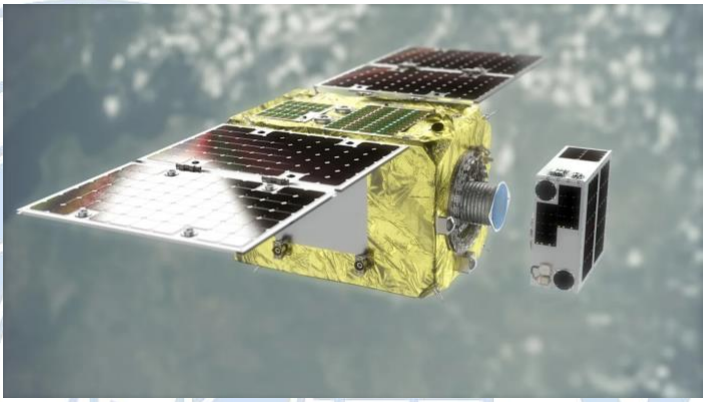

Figure 4: CG Image of ELSA-d Approaching Debris

Key Players

United States of America

USA is practically leading the deployment and operation of mega-constellations at LEO, as it holds unrivaled private companies such as Starlink and Project Kuiper. At the same time, through the U.S. Department of Defense (DoD), it directly possesses and controls the world's largest comprehensive space object database and Space Situational Awareness (SSA) tracking capabilities, possessing absolute influence and importance in global space traffic management.

The USA strongly favors non-binding soft law approaches, such as voluntary guidelines and best practices, over binding international regulations or orbital restriction treaties that hinder the innovation and growth of its commercial private entities. Consequently, while actively easing regulatory burdens that constrain commercial growth, it pursues policies to secure both national security and commercial leadership by designating the Department of Commerce (DOC) as the public interface to provide basic space safety data free of charge, while retaining high-precision military tracking data and the authoritative comprehensive catalog under the continued management of the DoD.

China

China is a major space power that has independently operated the space station "Tiangong" and pursued state-led mega-constellations (such as Guowang). In growing orbital friction with the U.S., China submitted an official diplomatic note (UN Document A/AC.105/1262) to the UN Secretary-General following a 2021 incident. Specifically, in the note, China reported that the Tiangong Space Station was forced to take two emergency "preventive collision avoidance maneuvers" on July 1 and October 21, 2021, because of Starlink satellites (Starlink-1095 and Starlink-2305). China is significantly relevant and influential as a key actor checking the U.S. and leading multilateral management in global space safety governance discussions.

China is calling for strict state responsibility for space activities and the establishment of a UN-centered, binding international law. It also clarifies that, based on OST Article VI, dangerous activities of the USA's private companies are also under the responsibility of the USA government. Furthermore, it holds the position of strictly defending space security and national sovereignty by promoting mandatory advance notifications and data sharing to prevent unilateral LEO monopolization.

Possible Solutions

Establishing a Centralized UN-Led Global Space Traffic Management (STM) Framework

To address the current deficits of international space law and the fragmented, military-led Space Situational Awareness (SSA) network, UN COPUOS should lead the establishment of a centralized and binding Global Space Traffic Management (STM) framework. The present system, operated by the U.S. Space Force or EU SST, depends on security-sensitive regional tracking databases and non-binding guidelines. To operate this, UN COPUOS would be the central body overseeing a standardized coordinate system, and designated technical entities would aggregate non-classified orbital data from governments and commercial operators. Additionally, national space regulatory agencies would enforce mandatory post-mission deorbit timeframes and automated collision-avoidance algorithms through their licensing processes, while UN COPUOS manages equitable LEO orbital allocation rules as binding multilateral protocols. Implementing such measures would prevent orbital monopolization and guarantee fair, non-discriminatory access to outer space for developing and non-spacefaring nations.

Establishing an UN-Backed Global Space Remediation Fund for Active Debris Removal

Active debris removal (ADR) missions currently face excessive financial and legal barriers. Public-private partnerships such as ClearSpace-1 and Astroscale have already demonstrated technological feasibility, but expanding this technology to routine operations requires a structured, multi-nation financing mechanism based on the international legal principle of common but differentiated responsibilities. Under this model, major space powers and commercial mega-constellation operators would contribute funding based on their debris contribution, which is measured through factors such as total deployed mass, orbital cross-sectional area, and historical debris generation. This fund would directly support commercial ADR contracts, promote private debris cleanup innovations, and effectively transform fragmented technological successes into a sustainable, global remediation framework.

Key Terms

Space Debris

Non-functional, artificial objects or fragments in Earth's orbit, ranging from defunct satellites and abandoned rocket bodies to untraceable micro-debris.

Soft law

Non-binding international legal instruments, such as voluntary guidelines, standards, and best practices, that lack legal enforceability but encourage compliance through diplomatic consensus.

Mega-Constellations

Large networks composed of hundreds to thousands of small satellites deployed primarily in Low Earth Orbit (LEO) to provide global, continuous services like high-speed internet.

Space Traffic Management (STM)

A unified framework of operational, technical, and regulatory measures designed to coordinate orbital movements, prevent collisions, and ensure safe, sustainable access to outer space.

Kessler Syndrome

A theoretical scenario where orbital collisions trigger a self-sustaining chain reaction of further debris, rendering specific orbits unusable.

Space Situational Awareness (SSA)

The monitoring and tracking of space objects and debris using sensors to predict potential orbital collisions.

Active Debris Removal (ADR)

Technologies used to capture and remove non-functional debris from orbit through physical or magnetic mechanisms.

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